AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM254
Managing IT: the why, the what and the how

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 10 /Apr./2025

Number of Exam Pages: 04 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **03** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

AOU EGYPT

CamScanner

PART 1: MCQ [10 Marks]

Each question carries a weight of 2 marks

Question 1: [10 marks] Read the following questions and choose the best answer.

1- Which of the following ITIL lifecycle stages is responsible for the day-to-day management of services, ensuring continuity in the face of both expected and unexpected events?
   A. Service Strategy
   B. Service Design
   C. Service Transition
   D. Service Operation

2- The key points about service validation and testing are that testing during service transition ensures that the services are:
   A. fit for purpose
   B. fit for use
   C. fit for purpose and fit for use
   D. None of the above

3- Which of the following metrics are part of Continual Service Improvement (CSI)?
   A. Technology metrics
   B. Process metrics
   C. Service metrics
   D. All of the above

4- Which of the following is focused on answering high-level strategic questions, such as 'What services should we offer, to whom, and why'?
   A. Strategy management for IT services
   B. Service portfolio management
   C. Financial management for IT services
   D. Demand management

5- Which of the following metrics is quantifiable and used to track progress toward a specific goal or objective?
   A. Critical Success Factor (CSF)
   B. Key Performance Indicator (KPI)
   C. Baselines
   D. Key Success Indicator (KSI

FOU EGYPT

CamScanner

PART 2: Short questions [30 Marks]

Question 2: [6 marks]

a) What is the purpose and scope of the service strategy phase in the service lifecycle? (3 marks)

b) List and define three types of service providers. (3 marks)

Question 3: [6 marks]

a) What is the role of Service Transition in the ITIL Service Lifecycle? (3 marks)

b) ITIL defines seven processes associated with the Service Transition stage of the service lifecycle. Briefly describe the purpose of the following processes: (3 marks)

- Service asset and configuration management
- Release and Deployment Management
- Change Evaluation

Question 4: [4 marks]

In the ITIL service lifecycle, service design includes several processes, activities, and roles. Briefly describe the purpose of the following:

- Service Catalogue Management
- Availability Management
- IT Service Continuity Management
- Information Security Management

Question 5: [4 marks]

List and define four essential characteristics that distinguish services from physical goods.

Question 6: [4 marks]

Requirements are concerned with the attributes of a system that enable it to fulfill its aims and objectives. List and describe four desirable characteristics of a good set of requirements.

Question 7: [3 marks]

A function is a team or group of people, along with the tools and resources they use to carry out one or more processes or activities. List and define three of the four main functions in Service Operation.

Question 8: [3 marks]

Briefly describe the differences between Operational Teams, Project Teams, and Collocated Teams.

AOU EGYPT

CamScanner

Part 3: Problem Solving Questions [20 Marks]

Question 9: [20 marks]

Worldwide Bank, a leading financial institution, provides online banking services to thousands of customers. The platform enables users to check balances, transfer funds, and make bill payments, and it complies with data privacy regulations established by the bank to safeguard customers' information. The bank also has a dedicated service desk team responsible for routine tasks such as password resets, user account management (creating, updating, and deleting accounts), software installations and updates, and responding to user inquiries.

On a Monday morning, customers began reporting that they were unable to log in to their accounts or complete transactions. Complaints quickly flooded the customer support center, social media platforms, and branch offices. Recognizing the widespread impact, the support team identified a disruption in the online banking service.

By 8:30 AM, the issue escalated to a critical level, prompting the service desk to classify it as a high-priority incident. After initial troubleshooting efforts failed, the case was escalated to the technical team. Upon investigation, they discovered that a server failure was causing the online banking outage, preventing customers from accessing their accounts and completing transactions. Within 30 minutes, the technical team manually rerouted traffic to the secondary server, successfully restoring service and resolving the incident.

Following the incident, the bank's management teams held an urgent meeting to determine the root cause of the failure. Given that the online banking service must be available 24/7, even minor disruptions were deemed unacceptable, especially considering that redundancy and backup mechanisms were in place.

Following the incident, the engineering team conducted a root analysis. Further analysis by the engineering team revealed that the secondary server had failed to take over immediately due to a power outage affecting its electricity source. To prevent future occurrences, the technical team installed backup generators and added a third server to enhance redundancy. These improvements were documented as part of the problem-resolution process.

a) What steps were taken in incident management to restore services? (3 marks)

b) How did problem management address the root cause of the outage? (3 marks)

c) How is request fulfillment managed in the bank? (3 marks)

d) Identify and define the Incident, Event, and Problem in this scenario. (6 marks)

e) Identify all the stakeholders involved in this service. (3 marks)

f) What are the functional, and compliance requirements of the online banking platform? (2 marks)

End of questions

CamScanner